

There have now been numerous experiments confirming this. The granular prefrontal cortex becomes uniquely active during tasks that require *self-reference*, such as evaluating your own personality traits, general self-related mind wandering, considering your own feelings, thinking about your own intentions, and thinking about yourself in general.

With this clue the granular prefrontal cortex is uniquely activated by self-reference, might we have missed some subtle but crucial impairments of granular prefrontal damage?

In 2015, scientists did the following study. They gave participants a neutral cue word (e.g., *bird* or *restaurant*) and asked them to tell the experimenter different narratives of *themselves* associated with that word. Some of these participants were healthy, some had damage to areas of the granular prefrontal cortex, and some had damage to the hippocampus.

How did people's narratives differ across these conditions? Humans with damage to the granular prefrontal areas but with the aPFC and hippocampus intact were able to imagine complex scenes, rich with detail, but they were impaired at imagining *themselves* in such scenes. They sometimes even completely omitted themselves from their narratives. Damage to the hippocampus seemed to have the opposite effect—patients could imagine themselves in past or future situations just fine but struggled to build external features of the world; they were unable to describe in detail any of the surrounding elements.

This suggests that the granular prefrontal cortex plays a key role in your ability to project *yourself*—your intentions, feelings, thoughts, personality, and knowledge—into your rendered simulations, whether they are about the past or some imagined future. The simulations run in rat brains, which has no gPFC and only an aPFC and hippocampus, show evidence of rendering an external world, but there is nothing to suggest that they truly project any meaningful model of *themselves* into these simulations.

Damage to the granular prefrontal cortex may also affect people's models of themselves not only in their mental simulations, but also in the *present*. Some people with gPFC damage develop mirror-sign syndrome, a condition in which they no longer recognize themselves in a mirror. These patients *insist* that the people they see in a mirror are not *them*. Your modeling of your own mind in the present and the mind you project in your imagination seem to be intimately related.

The idea that the new primate areas take part in modeling *your own mind* makes sense when you follow their input/output connectivity. The older mammalian aPFC gets input directly from the amygdala and hippocampus, while the new primate gPFC receives almost no amygdala or hippocampal input or any direct sensory input at all. Instead, the primate gPFC gets most of its input directly from the older aPFC.

One interpretation of this is that these new primate areas are constructing a generative model of the older mammalian aPFC and sensory cortex itself. Just as aPFC constructs explanations of amygdala and hippocampus activity (invents “intent”), perhaps the gPFC constructs explanations of the aPFC’s model of intent—possibly inventing what one might call *a mind*. Perhaps the gPFC and PSC construct a model of one’s own inner simulation to explain one’s intentions in the aPFC given knowledge in the sensory neocortex.

Let’s use a thought experiment to build some intuition about what this means. Suppose you put our ancestral primate in a maze. When it reached a choice point, it turned left. Suppose you could ask its different brain areas why the animal turned left. You would get very different answers at each level of abstraction. Reflexes would say, *Because I have an evolutionarily hard-coded rule to turn toward the smell coming from the left*. Vertebrate structures would say, *Because going left maximizes predicted future reward*. Mammalian structures would say, *Because left leads to food*. But primate structures would say, *Because I’m hungry, eating feels good when I am hungry, and to the best of my knowledge, going left leads to food*. In other words, the gPFC constructs explanations of the simulation itself, of what the animal wants and knows and thinks. Psychologists and philosophers call this *metacognition*: the ability to think about thinking.

What mammals find in their inner simulations of the external world is, in some sense, the same thing as their knowledge about the external world. When a mammal simulates going down a path and its sensory neocortex renders a simulation that contains water at the end of the path, this is the same thing as “knowing” that water exists at the end of the path. While the older mammalian areas of the sensory neocortex render the simulation of the external world (containing knowledge), the new primate areas of neocortex (what I have been calling the primate sensory cortex) seems to create a model of this knowledge itself (areas of PSC get input from various areas of sensory neocortex). These new primate areas try to explain *why* the

sensory neocortex believes food is over there, why an animal's inner simulation of the external world is the way it is. An answer might be: *Because last time I went over there I saw water, and hence when I simulate going back there, I see water in my imagination.* Put slightly differently: *Because I last saw water there, I now know that water is over there even though before I did not.*

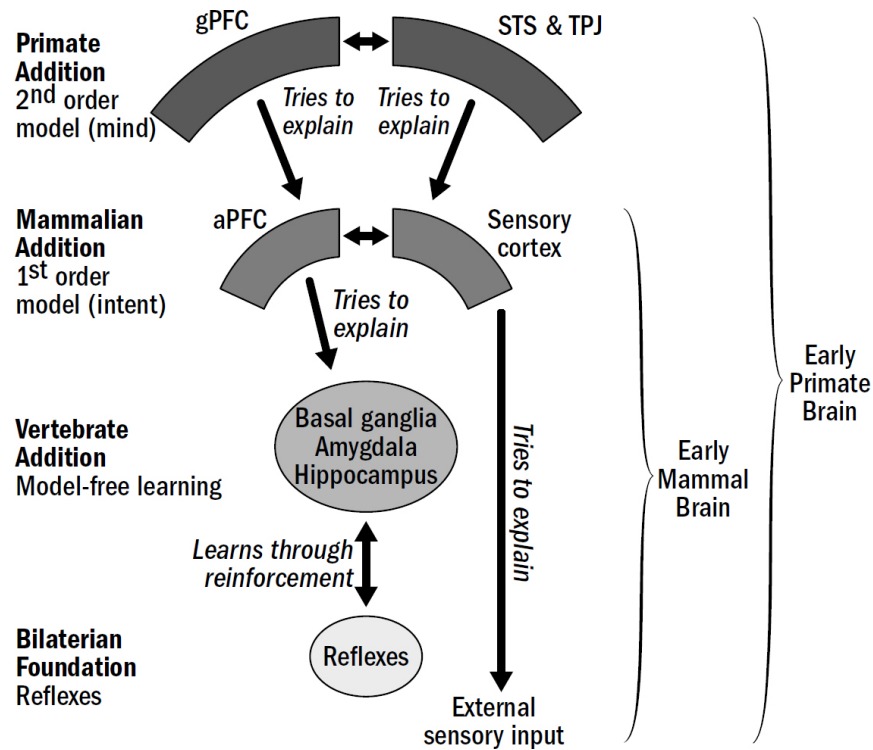


Figure 16.3

Original art by Rebecca Gelernter

These systems are all bootstrapped on one another. Reflexes drive valence responses without any learning required, making choices based on evolutionarily hard-coded rules. The vertebrate basal ganglia and amygdala can *then* learn new behaviors based on what has historically been reinforced by these reflexes, making choices based on maximizing reward. The mammalian aPFC can then learn a generative model of this model-free behavior and construct explanations, making choices based on imagined goals (e.g., drinking water). This could be considered a first-order model. The primate gPFC can then learn a more abstract generative model (a second-order model) of this aPFC-driven behavior and construct